(Please write your Enrollment Number)

MID-TERM EXAMINATION
Course : MAE & DMAM (Semester 3)
(October, 2023) OFF LINE mode

Subject Code: BMA-201 | **Subject: Production Technology I**

Time: 1 ½Hours | **Maximum Marks : 30**

Note: Q. 1 is compulsory.

Q1		**(2.5*4)**	
	(a) Consider that you have been employed as a GET in a company making sink taps. The company uses sand casting technique, employ Core and Chaplets. Brief about importance of core and chaplets. Make a sketch to show use of chaplets.		
	(b) Discuss desirable properties of molding sand, if you have to cast Aluminum.		
	(c) What are the most common welding defects in Shielded Metal Arc Welding process? Consider you are using DC machine.		
	(d) Consider that you are a trainer in your organization and you have to discuss about importance of coating on Electrodes in case of stick welding (SMAW) process. How will you explain?		
Q2	**(Attempt any Two Parts) UNIT-1**	**(5,5)**	
	(a) What is sand casting? Draw a sketch to show different features in a sand mold. Explain in points the sand mold casting process.		
	(b) Differentiate between true centrifugal and semi-centrifugal casting processes.		
	(c) What is a pattern in Casting? Explain any four types of patterns with neat sketches that may be used in sand casting.		
Q3	***(Attempt any Two Parts) UNIT-2***	**(5,5)**	
	(a) What are the various tools and equipment used in Oxy Acetylene welding? Write in brief about different types of flames with neat sketches.		
	(b) Define Submerged arc welding process with neat sketch. What are its advantages and disadvantages?		
	(c) Differentiate between electron beam welding and laser beam welding.		